INCH-POUND

MIL-PRF-81757/17(AS)
10 July 2009
SUPERSEDING
MIL-B-8565/10(AS)
14 June 1993

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, HIGH-RATE, TYPE 2, 24-VOLT, 35-AMPERE-HOUR, CHARGE CONTROLLED

This specification is approved for use by Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81757.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the batteries covered by this specification sheet shall be D8565/10-1. The removable sensor/connector assembly PIN shall be D8565/10-2. The D8565/10-1 shall be a limited-repair battery. The D8565/10-2 sensor/connector assembly shall not be repairable.

2. Relationship among PINS. The removable sensor/connector assembly PIN D8565/10-2 shall be installed as an integral part of PIN D8565/10-1. PIN D8565/10-2 may be purchased as an individual item separately from the PIN D8565/10-1 battery.

3. Dimensions and weights. The dimensions and configuration of the battery container and container cover shall be as specified in 3.10.8, as modified by this specification sheet and as shown on figure 1 of this specification sheet. The weight of the battery shall be not greater than 85.0 pounds (38.64 kilograms) (see 3.10.8 and 4.5.8 of MIL-PRF-81757).

4. Capacity. The capacity of the battery shall be not less than 35 ampere-hours (Ah). See 3.10.14 as modified herein.

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5. Interface requirements. The battery shall contain 19 cells. The cells, the D8565/10-2 sensor/connector assembly, and other interior components of the battery shall be arranged and positioned as shown on figure 2 of this specification sheet. The fasteners specified on figure 2 are required due to interfaces with the MS3509-32 receptacle and the D8565/10-2 harness sensor assembly.

6. Interface requirements between the battery and the sensor assembly. The partition that retains the center row of cells shall intimately contact the fourth cell from the positive terminal of the battery. No space or air passage shall be allowed at this interface. The partition shall be the same height ± 0.1 inch as the plates within the cells that it retains. The partition shall be rigid and strong enough to contain the cells for the life of the battery without warping or breaking.

6.1 Mount. The partition shall have three 0.25-inch long, size 6-32 studs permanently affixed. The material of the studs shall meet the requirements of 3.10.2 of MIL-PRF-81757 and 3.10.15 and 3.10.19 as modified herein (see 6.10.2.2 of MIL-PRF-81757/17). The studs shall be equally spaced 120 ± 1º apart at the 12-, 4-, and 8-o'clock positions on a 2.00 ± 0.05 inch diameter circle. The center of the circle and the center of the mounting hole of the J2 receptacle shall be aligned on the same horizontal axis within 0.06 inch.

6.2. Assembly interface. The sensor/connector assembly PIN D8565/10-2 shall be installed in each supplied battery PIN D8565/10-1. The temperature sensing subassembly of PIN D8565/10-2 shall be mounted on the battery partition by use of the three studs specified herein. A size 6-32, insulated insert locking nut shall be installed on each stud to fasten the subassembly. The material of the nut shall meet the requirements of 3.10.2, 3.10.3, and 3.10.4 of MIL-PRF-81757 and 3.10.15 and 3.10.19 as modified herein (see 6.10.2.2 of MIL-PRF-81757/17).

7. Assembly design. The sensor/connector assembly PIN D8565/10-2 shall comply with figures 3 and 4 of this specification sheet. This entire assembly consists of the following: (1) The temperature sensing subassembly; (2) The connector receptacle J2; (3) Four electrical wires that interconnect the temperature sensing subassembly and J2; and (4) One electrical wire attached to J2 and the other end attached to a ring terminal connector. Three locking nuts specified in 6.2 contained herein to be used for mounting the assembly shall be included with each PIN D8565/10-2. This assembly shall comply with the following subparagraphs as specified herein. Assemblies shall be interchangeable regardless of battery manufacturer.

7.1 J2 receptacle. The J2 connector receptacle shall conform to MS3474W10-6P and shall mate with a connector plug conforming to MS3476L10-6S or equal. It shall be located as shown on figure 1 and electrically connected as shown on figures 3 and 4. The shell shall be constructed of a material that shall meet the requirements of 3.10.2 of MIL-PRF-81757 and 3.10.15 and 3.10.19 as modified herein (see 6.10.2.2 of MIL-PRF-81757/17). The shell shall not be plated. The connector receptacle need not be hermetically sealed.

7.2 Electrical wires. All electrical lead wires shall be American Wire Gage (AWG) number 20 (19X32 stranding) and shall be plated with conductive material that will protect the substrate from the electrolyte (see 6.10.2.2 as modified herein).

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7.3 Potting. The electrical wire connections at the J2 receptacle shall be potted with an insulating compound that meets the requirements of 3.10.3 of MIL-PRF-81757 (see 6.10.2.2 of MIL-PRF-81757/17).

7.4 Wiring. The length of electrical wire for connecting pin A of receptacle J2 to the battery's center-tap voltage (shown on figures 2 and 3) shall be 15.5 ± 0.5 inches. This length shall exclude the connections at both ends. An electrical connector shall be attached to the free end for attaching to the battery's center-tap voltage connection. Soldering or brazing shall not be used to attach the electrical connector to the wire of pin A of receptacle J2. The connector shall be equivalent to a ring terminal connector that has an insulating sleeve and a size 6 screw hole. The material of the connector shall meet the requirements of 3.10.2, 3.10.3, 3.10.4, of MIL-PRF-81757 and 3.10.15, and 3.10.19 as modified by this specification sheet (see 6.10.2.2 of MIL-PRF-81757/17).

7.5 Wire length. The length of each of the electrical wires that connect pins B, C, D, and E of receptacle J2 to the temperature sensing subassembly (shown on figures 2 and 3) shall be 6.0 ± 1.0 inches. This length shall exclude the fastened connections at both ends of the electrical wire.

7.6 Components. The temperature transducer shall be equivalent to Analog Devices, Inc. part number AD590L. The capacitor shall be equivalent to M39014/01-1473 or M39014/02-1230. The temperature-activated switch shall be equivalent to Texas Instruments, Inc. type M2S. The switch contacts shall be normally open. These items are required due to interchangeability requirements (see 7 above) and to interface requirements with aircraft temperature sensing equipment and aircraft electromagnetic interference (EMI) requirements.

7.7 Temperature sensing subassembly. The temperature sensing subassembly shall consist of the temperature transducer with the capacitor specified in 7.6 of this specification sheet, the temperature switch specified in 7.6, and a mounting plate containing these items as shown on figure 3 of this specification sheet.

7.7.1. Mounting plate characteristics. The mounting plate shall be thermally conductive, electrically conductive, contain no plating, and meet the requirements of 3.10.2 of MIL-PRF-81757, and 3.10.15 and 3.10.19 as modified by this specification sheet (see 6.10.2.2 of MIL-PRF-81757/17.)

7.7.2. Contact requirement. The contact surfaces between the temperature transducer and the mounting plate shall be intimate to allow thermal conductivity.

7.7.3. Location. The temperature transducer and the temperature-activated switch shall be located near the center of the plate. They shall be potted inside a container that shall be secured to the mounting plate.

7.7.4. Container requirements. The container for the transducer and switch shall be capable of being welded to the mounting plate. The container shall meet the requirements of 3.10.2 of MIL-PRF-81757, and 3.10.15 and 3.10.19 as modified by this specification sheet (see 6.10.2.2 of

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MIL-PRF-81757/17). The exterior dimensions of the potting cavity shall be as shown on figure 3 of this specification sheet.

7.7.5. Flange. The potting cavity shall be centered on the plate so that the plate forms a mounting flange as shown on figure 3 of this specification sheet. The plate's mounting flange shall contain three slotted holes shown on figure 3 of this specification sheet for mounting the subassembly to the D8565/10-1 battery partition.

7.8 Assembly weight. The weight of PIN D8565/10-2 shall be not greater than 0.4 pound (0.18 kilogram).

7.9 Mounting holes. The three vertically slotted holes shown and specified on figure 3 of this specification sheet shall be located and sized to permit the D8565/10-2 assembly to be mounted on the three 6-32 studs on the partition of the D8565/10-1 battery.

7.10 Markings. The sensor/connector assembly shall contain identification markings that show the following: military part or identifying number (PIN), national stock number (NSN), contractor's Commercial and Government Entity (CAGE) code or manufacturer's name, and contract number. The markings shall be located as shown on figure 1.

7.1l. Performance. When the assembly is an integral part of the purchased battery, the assembly shall withstand without damage or degraded performance all environmental tests to which the battery is subjected. Examples of damage include cracking, chipping, blowholes, and degradation of material.

8. Table I. Table I replaces table III of MIL-PRF-81757 and lists the required inspections that ensure the quality of the battery with the sensor/connector assembly installed.

9. Table II. Table II lists the required inspections that ensure the quality of assembly PIN D8565/10-2 when purchased as an individual item separately from battery PIN D8565/10-1.

10. Compliance. PINs D8565/10-1 and D8565/10-2 shall comply with MIL-PRF-81757 except as follows:

10.1 The following paragraphs of MIL-PRF-81757 are not applicable:

3.10.14.1 Initial capacity.
3.10.18 Altitude.
3.10.20 Twenty-second pulse discharge.
3.10.21 Medium-rate discharge, operating position.
3.10.29 Storage effects.
4.5.18 Altitude test.
4.5.20 Twenty-second pulse discharge test.
4.5.21 Medium rate discharge, operating position test.
4.5.29 Storage effects test.

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10.2 Modify the following paragraphs of MIL-PRF-81757:

2.2.1 Specifications and standards. Add the following documents:

SPECIFICATIONS:

"MIL-PRF-5606 -	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance. Inactive for new design"
"MIL-PRF-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number O-156"
"MIL-PRF-39014/1 -	Capacitors, Fixed, Ceramic Dielectric (General Purpose), Established Reliability and Nonestablished Reliability, Style CKR05, NATO Type Designation NCC61"
"MIL-PRF-39014/2 -	Capacitors, Fixed, Ceramic Dielectric (General Purpose), Established reliability and Nonestablished Reliability, Style CKR06, NATO Type Designation NCC62"
"MIL-PRF-85570 -	Cleaning Compounds, Aircraft, Exterior"
"MIL-DTL-26482 -	Connectors, Electrical, (Circular, Miniature, Quick Disconnect, Environment Resisting), Receptacles and Plugs, General Specification for"

STANDARDS:

"MS3474	Connectors, Receptacle, Electrical, Series 2, Crimp Type, Rear Mounting Jam Nut, Bayonet Coupling, Classes A, D, L, T, W, and Z
"MS3476	Connectors, Plug, Electric, Series 2, Crimp Type, Bayonet Coupling, Classes A, D, L, T, W, and Z"
"MS3509	Receptacles, Electric, Aircraft Storage Battery"

3.4 Materials and components. Delete the second sentence and insert: "Aluminum, polycarbonate, brass, or bronze (see 6.10.2.2 of MIL-PRF-81757/17) shall not be used in the construction of any battery, cell, or component covered by this specification."

3.7.2 Battery container and cover. Add after the last sentence: "The elastomer gasket of the container cover shall have elastomer projections that contact each intercell connector. The projections shall provide sufficient pressure to restrict vertical movement of the cells during the vibration test of 4.5.16 as modified herein."

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3.7.5 Handles. Add as a new last sentence: "The container shall contain two identical handles equivalent to either part number 980-SS-2-09 or 955-SS-2 manufactured by Nielsen Hardware, Zero Corporation, Hartford, CT 06141."

3.7.8 Receptacles. Following the last sentence, add: "The J1 receptacle shall conform to MS3509-32. It shall be located so that the contact surfaces of each terminal block are in the same plane as the contact surfaces of the interconnecting cell terminal."

3.7.11.8 Electrolyte. Delete the first two sentences and insert: "Each cell shall contain the amount of electrolyte that will provide a level between 0.375 and 0.50 inch above the bottom of the baffle immediately after completing the charge in accordance with 4.4.3 as modified herein and without adding any electrolyte. The electrolyte chosen for use shall enable the battery to meet the requirements specified herein (see 6.10.5 of MIL-PRF-81757). Use of any expanders or anti-agglomerate substances equivalent to cellulose oxide shall be approved by the cognizant Government engineering activity before cell manufacture."

3.10.8 Dimensions and weights. Delete in its entirety and substitute: “The dimensions and configuration of the battery container, container cover, and sensor assembly shall be as shown on figures 1, 2, 3, and 4 of the specification sheet. The weight of the battery shall not be greater than 86.0 pounds (39 kilograms) with the electrolyte adjusted in accordance with 4.4.4, as modified by MIL-PRF-81757/17.”

3.10.9 Bulging. Following the last sentence, add: "The partition that holds the temperature sensing subassembly of PIN D8565/10-2 shall not distort or bulge and remain within the dimensions of figure 3 of MIL-PRF-81757/17 when subjected to any test in MIL-PRF-81757 or MIL-PRF-81757/17 (see 3.10.9 of MIL-PRF-81757)."

3.10.14 Capacity. Delete the text and insert: "The minimum capacity (see 6.9, 6.13.1, 6.13.2, and 6.13.14 all of MIL-PRF-81757) shall be 35 ampere-hours (Ah). The battery shall provide not less than 35 Ah capacity during the final discharge of the life cycle test (see 4.5.23 of MIL-PRF-81757). This shall be based on a constant 35-ampere discharge at room temperature to an end voltage of 18.0 volts, and the discharge time shall be not less than 1.0 hour."

3.10.15 Humidity and charge retention. In subparagraph a(2), change "45 minutes" to "54 minutes".

3.10.19 Salt fog. Delete the last sentence and insert: "The battery shall also meet the requirements of 3.9 of MIL-PRF-81757 and 3.10.34 as specified herein."

3.10.22 Constant voltage discharge. Delete the first sentence and insert: "The battery shall produce not less than 1400 amperes (24º ± 2ºC) and 650 amperes (-26º ± 2ºC) at 15 seconds into the test after being tested in accordance with 4.5.22 of MIL-PRF-81757.”

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4.4.2 Constant potential charging method. In subparagraph b, delete: "1.5 volts per cell" and substitute: "28.25 ± 0.10 volts". Following the first sentence, add: "The voltage may fall outside this range for the first 5 minutes."

4.4.3 Constant current charging method. Delete the third sentence and insert: "Constant current charge the battery at 35 amperes to 29.5 ± 0.1 volts and then reduce the charge current to 7 amperes. Continue the charge at 7 amperes for 3.0 ± 0.1 hours."

4.4.4 Electrolyte level adjustment. Delete the text and insert: "At the end of charge, adjust the electrolyte in each cell to a level between one-fourth and one-half inch above the bottom of the cell baffle by adding deionized or distilled water."

4.5.2 Electrolyte soak of metal parts. Delete the text and insert: “Subject samples of all metal parts of the batteries, cells, and components to this test. Additionally, a panel with dimensions 4 inches by 6 inches shall be furnished for this test, coated as it would be supplied on the battery container and containing sample identification markings. Each sample shall be subjected to each test solution as indicated in the following steps:

“a. For the battery container, the sensor assembly (except for that portion of J2 receptacle that is outside the battery container), and the J1 receptacle: Soak one-half of each specimen along its horizontal length in an aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity and allow to stand for 168 ± 2 hours at 65.5° ± 1.1 °C (150°± 2 °F). Rinse, dry, and examine the specimens for the requirements of 3.10.2 of MIL-PRF-81757.

“b. For the battery container, battery cover, all markings, handles, latches, and the J1 and J2 receptacles, the test solutions and periods are:

“1. An aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity for 4 ± 0.2 hours. Blue and black receptacles shall be tested for 168 hours at 66° ± 2° C (150° ± 3.6° F).

“2. A 50 percent aqueous solution of ethylene glycol for 20 ± 1 hours.

“3. MIL-PRF-5606 aircraft hydraulic fluid for 20 ± 1 hours. Note: Do not perform step 4.5.2.c.2.

“4. MIL-PRF-23699 aircraft engine lubricating oil for 20 ± 1 hours. Note: Do not perform step 4.5.2.c.2.

“5. MIL-PRF-85570 Type I exterior aircraft cleaner, 1 part cleaner to 10 parts tap water, for 20 ± 1 hours.

“c. This test for the solutions of 4.5.2.b shall consist of the following steps.

“1. Measure the major dimensions of each specimen with a micrometer. Weigh each specimen on a chemical balance. Place the specimen in a vessel with a lid.

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Completely immerse the specimen in the room temperature test solution, cover the vessel, and allow the vessel to stand at room temperature for the indicated length of time (see 4.4.1 of MIL-PRF-81757).

"2. Rinse the specimen in water. However, do not rinse the specimens of 4.5.2.b.3 and 4.5.2.b.4.

"3. Wipe dry. Examine the identification marking. Inspect all samples for evidence of cracks or blisters. Measure and weigh the samples. Calculate the percentage change in major dimensions and weight. Examine the item for the requirements of 3.10.34 as specified herein.

"4. Repeat steps 4.5.2.c.1 through 4.5.2.c.3 for each of the indicated test solutions of 4.5.2.b in turn."

4.5.16 Vibration test. Delete the sixth and seventh sentence and insert: "Subject the battery to a random vibration shown on figure 7 of this specification sheet along each of three mutually perpendicular axes. The vibration time in each axis shall be two (2) hours." Delete the ninth sentence and insert: "discharge the battery at 3.5 amperes during vibration." Delete the twelfth sentence and insert: "discharge the battery in accordance with 4.5.22 of MIL-PRF-81757. Add to the end of the last sentence: "and 3.10.31 of MIL-PRF-81757/17.".

4.5.25 Physical integrity at high temperature (85ºC). Subparagraph c delete the text and substitute: "Remove the battery from the 85°C (185°F) environment. Lift the battery by its handles. Place the battery on a flat surface. Observe any evidence of distortion of the case."

6.2 Acquisition requirements. Insert a new subparagraph:

"i. Units must be shipped only after verification testing is completed and the testing agency has authorized release of the shipment."

6.10.1 Corrosive nature of potassium hydroxide. Delete the fourth sentence and insert: "Potassium hydroxide attacks aluminum, glass, many plastics, rubber compounds, brass, tin, zinc, lead, bronze, and the corrosion preventive compound used on batteries (see 3.4 and 3.7.2 as modified herein, 3.7.3 and 3.7.12 of MIL-PRF-81757 and requirements 6 and 7 of MIL-PRF-81757/17)."

6.10.2 Elastomers and potassium hydroxide. Change the paragraph number to 6.10.2.1. Delete "and potassium hydroxide" from the title.

FIGURE 1. Marking for limited-repair batteries. Delete the "Initial Capacity" line and substitute: "INITIAL CAPACITY: 35 AH (35 AMPS/60 MINUTES/24ºC/18 V)." Delete the "Minimum Capacity" line.

10.3 Add the following paragraphs:

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"3.10.31 Vibration. The battery shall meet the requirements of 3.7.2 as modified herein and 3.9 of MIL-PRF-81757 when subjected to the vibration test of 4.5.16 as modified herein."

"3.10.32 Profile discharge at -18ºC. The battery shall produce not less than the minimum voltage requirements shown on figure 5 of the specification sheet (see 3.1 of MIL-PRF-81757) for two of the three profile discharges conducted after being tested in accordance with 4.5.32 of MIL-PRF-81757/17."

"3.10.33 Sensor/connector assembly.

"3.10.33.1 Harness alkali resistance. External and internal metal parts of each battery, cell, and component shall not show any of the deleterious effects listed in table III during or after the testing specified in 4.5.33.1 of MIL-PRF-81757/17. The battery shall also meet the requirements of 3.9 of MIL-PRF-81757.

"3.10.33.2 Temperature accuracy. The accuracy of the assembly shall be ±3°C (± 5ºF) as determined after being tested in accordance with 4.5.33.2 of MIL-PRF-81757/17.

"3.10.33.3 Switch activation. The temperature switch contacts of the assembly shall close at 74 ± 3ºC (165 ± 5°F) on a temperature rise and reopen during a temperature drop after being tested in accordance with 4.5.33.3 of MIL-PRF-81757/17. The temperature differential between closure and reopening of the contacts shall be not greater than 6ºC (11°F).

"3.10.33.4 Switch contact resistance. When the contacts of the temperature-activated switch are in the open position, the resistance between the contacts shall be not less than 1.0 million ohms after being tested in accordance with 4.5.33.4 of MIL-PRF-81757/17.

"3.10.33.5 Vibration. The assembly shall meet the requirements of 3.10.33.2, 3.10.33.3, 3.10.33.4 of MIL-PRF-81757/17, and 3.9 of MIL-PRF-81757 after being tested in accordance with 4.5.33.5 of MIL-PRF-81757/17. The assembly shall show no mechanical failure or damage.

“3.10.34 Immersion and salt fog resistance. The connector shall not exhibit deleterious effects indicated below during or after the tests of 4.5.2 as modified herein, 4.5.3 and 4.5.19 of MIL-PRF-81757. The connector shall also meet the requirements of 3.9 during and after testing. The plug shall also meet the requirements of 3.10.33.1, 3.10.33.2, and 3.10.33.3 of MIL-PRF-81757/17. The receptacle shall also meet the requirements of 3.10.33.1 and 3.10.33.4 of MIL-PRF-81757/17.”

“3.10.35 Corrosion resistance. External and internal metal or component parts of each battery shal1 show no evidence of cracking, pitting, chipping, sealing, corrosion, or other deleterious effects during the performance of any of the tests specified in 4.5 of MIL-PRF-81757 and MIL-PRF-81757/17.”

“3.10.36 Strength of receptacle. The receptacle and receptacle mounting shall withstand a pullout force of 250 ±25 pounds (1112 ±111 Newtons) and a torsional force of 6 ±0.5 foot-

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pounds (8.1 ±0.7 Newton-meters) when tested in accordance with 4.5.36 of MIL-PRF-81757/17."

"4.4.7 Battery temperature measurements. When the battery temperature will be measured during laboratory testing, the temperature shall be measured at the intersection of the four cells shown on figure 2 herein. The temperature shall be measured with a temperature sensor (thermocouple or thermistor) inserted into the intersection to a depth equal to the top of the cell plate stack."

"4.5.32 Profile discharge at -18ºC. This test shall consist of the following steps.

"a. Constant current charge the battery in accordance with 4.4.3 as modified in the specification sheet but do not adjust the electrolyte levels.

"b. Stabilize the battery at -18º ± 1.1ºC (0º ± 2ºF) for not less than 24 hours. The battery shall remain in the -18ºC environment for the duration of steps c, d, and e.

"c. Discharge the battery at 35 amperes for 2 minutes and then initiate the profile discharge of figure 5 of the specification sheet.

"d. After the profile discharge, rest the battery at 0.0 ampere for 4 minutes and then repeat the profile discharge of figure 5 of the specification sheet one time.

"e. Repeat step d one time.

"f. Examine the battery for the requirements of 3.10.32 of MIL-PRF-81757/17."

"4.5.33 Sensor/connector assembly tests. Notes: (1) Unless otherwise specified, the assembly shall be an integral part of battery PIN D8565/10-1 when conducting the tests of the following subparagraphs; (2) The electrical circuit shown on figure 6 of the specification sheet shall be used for conducting 4.5.33.2, 4.5.33.3, and 4.5.33.4 of MIL-PRF-81757/17; and (3) Prior to conducting the switch activation test of 4.5.33.3, the assembly shall comply with 3.10.33.2 of MIL-PRF-81757/17.

"4.5.33.1 Harness alkali resistance test. Detach the assembly from the battery and install it on the inside of a container which is capable of withstanding 1.300 ±0.025 specific gravity potassium hydroxide (KOH) electrolyte. Attach the harness connector to the container in the same way it is attached to the battery. Fill the container with electrolyte to cover all parts of the harness. Place the container in a 155ºF environment to bring the electrolyte solution to 155ºF. Soak the harness in the electrolyte at 155ºF for 168.0 ± 1.0 hours. Examine the assembly for the requirements of 3.10.33.1 of MIL-PRF-81757/17.

"4.5.33.2 Temperature accuracy test. Determine the ambient Fahrenheit temperature of the sensor as installed on the battery. This may be accomplished with the use of a calibrated glass stem thermometer. While the sensor is at this temperature, measure the voltage differential across resistor R1 with the use of a calibrated digital voltmeter (DVM) as shown on figure 6.

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Apply this measured differential voltage (ΔV) in a formula to calculate a temperature in degrees Fahrenheit (TºF). The formula for this calculation is T°C = (ΔV-2.554) X 100 – 17.78 or TºF = (ΔV - 2.554) X 180] [e.g., 77.0ºF = (2.982 volts - 2.554) X 180]. Compare the calculated temperature (TºF) with the determined sensor temperature. Examine the assembly for the requirements of 3.10.33.2 of MIL-PRF-81757/17.

"4.5.33.3 Switch activation test. Raise the temperature of the assembly to a point not greater than 77ºC and determine the temperature at which the switch contacts close. When at this raised temperature, subject the sensor to a temperature drop. Determine the temperature at which the switch contacts open. Examine the assembly for the requirements of 3.10.33.3 of MIL-PRF-81757/17.

"4.5.33.4 Switch contact resistance test. The temperature-activated switch contacts shall be in the open position when performing this test. Measure the resistance between pins D and E of connector J2. Examine the assembly for the requirements of 3.10.33.4 of MIL-PRF-81757/17.

"4.5.33.5 Vibration. Subject the assembly to 4.5.16 as modified in the specification sheet except the battery shall be open circuit. Afterwards, subject the assembly to 4.5.33.2, 4.5.33.3, and 4.5.33.4 of MIL-PRF-81757/17. Examine the assembly for the requirements of 3.10.33.5 of MIL-PRF-81757/17.

"4.5.33.6 Examination. Examine the detached assembly for compliance with the materials, components, design, construction, markings, and workmanship specified in requirements 6 and 7 of the specification sheet.

"4.5.33.7 Dimensions and weight. Examine the detached assembly to determine compliance with the dimensions and weight specified in 7.8 of the specification sheet.

"4.5.33.8 Other environmental. When the assembly is an integral part of the purchased battery, the assembly shall accompany the battery when subjected to environmental tests. The assembly shall comply with 6 and 7 of the specification sheet."

"4.5.34 Capacity Discharge. Stabilize the battery at room temperature (see 4.4.1 of MIL-PRF-81757). Discharge the battery at the 1.0 C-rate to a cutoff voltage (see 6.13.4 of MIL-PRF-81757) of 18.0 volts. Afterwards, for qualification inspection, constant potential charge the battery in accordance with 4.4.2 as modified herein. For group A conformance inspection, any charge method (from 4.4.2 and 4.4.3 both as modified herein, or other method not listed in section 4) may be selected as long as all batteries are charged by the same method and meet all the requirements of this specification. Examine the battery for the requirements of 3.10.14, as modified herein."

"4.5.35 Corrosion resistance. Samples of all metal parts of the cells and/or battery shall comply with 3.6.1 of MIL-PRF-81757 after being subjected to this test. A four by six-inch panel of battery case material shall be furnished for this test, coated as it would be supplied on the battery and containing sample identification markings. One-half of each specimen shall be immersed along its horizontal length in a solution of 1.300 +0.025 specific gravity KOH

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electrolyte and allowed to stand for 7 days at 65.5° +1.1°C (150° ±2°F). Specimens shal1 then be rinsed, dried and observed for the requirements of 3.10.35 of MIL-PRF-81757/17."

"4.5.36 Strength of receptacle. Insert a grooved steel shaft that mates with the receptacle socket such that the socket pins are secured in the steel shaft. Exert a tensional force of 250 ±25 pounds (1112 ±111 Newtons) against the socket pins. Following this test, subject the steel shaft to a torsional force of 6 ±0.5 foot-pounds (8.1 ±0.7 Newton-meters). Examine the receptacle mounting for the requirements of 3.10.36 of MIL-PRF-81757/17."

"4.5.38 Final examination. Examine the battery for the requirements of 3.9 of MIL-PRF-81757. Subject the battery to the dielectric strength test of 4.5.27 of MIL-PRF-81757. Examine the battery for the requirements of 3.10.27 of MIL-PRF-81757."

"6.10.2 Battery materials and potassium hydroxide."

"6.10.2.2 Other battery materials. Batteries that have passed the testing required for qualification and have worked acceptably in practice have used the following materials:

Tellurium copper (adapter blocks)
Stainless steel (ring terminal connector on free end of electrical wire attached to pin A of receptacle J2)
Stainless steel 300 series (container for temperature transducer and temperature-activated switch, battery case, sensor/connector assembly mounting plate and locking nut, partition studs, shell of J2 connector receptacle
Monel (shell of J2 connector receptacle)
Nickel-alloy material such as inconel (shell of J2 connector receptacle)
Nylon (insulating insert of locking nut, insulating insert of ring terminal connector on free end of electrical wire attached to pin A of receptacle J2)
Nickel-plated copper (Sensor/connector assembly electrical lead wires, ring terminal connector on free end of electrical wire attached to pin A of receptacle J2)
Silver-plated copper (Sensor/connector assembly electrical lead wires)
MIL-I-16923 (STYCAST 2651) (insulating compound for electrical wire connections of the sensor/connector assembly)"

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TABLE I. Qualification and quality conformance inspection of PIN D8565/10-1.

Test Number	Examination and tests 6/	Qualification inspection					Conformance inspection				
		Sample Number			Requirements paragraph 1/	Test paragraph 1/	Group A sample 100%	Group B sample		Group C sample	
		1	2	3				1	2	1	2
1	INCOMING INSPECTION	X	X	X	3.10.1(G)	4.5.1(G)	X			X	X
2	VISUAL AND MECHANICAL EXAMINATION	X	X	X	3.6(G), 3.7(G), 3.8(G), 3.10.6(G)	4.5.6(G)	X			X	X
3	CELL VENT TEST	X	X	X	3.10.13(G)	4.5.12(G)				X	X
4	DIMENSIONS AND WEIGHT	X		X	Reqt (3)(N), 3.10.8(G)	4.5.8(G)	X2/			X2/	X2/
5	DIELECTRIC STRENGTH AND INSULATION RESISTANCE	X	X	X	3.10.27(G)	4.5.27(G)	X				
6	COVER SEAL TEST	X	X	X	3.9(G), 3.10.11(G)	4.5.10(G)		X		X	X
7	HANDLES	X	X	X	3.9(G), 3.10.10(G)	4.5.9(G)				X	
8	STRENGTH OF RECEPTACLE	X		X	3.10.36(N)	4.5.36(N)					
9	VENTED FILLER CAP SEAL	X	X	X	3.7.12(G), 3.8.6(G), 3.9(G), 3.10.5(G),	4.5.5(G)	X			X	X
10	TEMPERATURE ACCURACY	X	X	X	3.10.33.2(N)	4.5.33.2(N)	X				
11	SWITCH ACTIVATION	X	X	X	3.10.33.3(N)	4.5.33.3(N)	X				
12	SWITCH CONTACT RESISTANCE	X	X	X	3.10.33.4(N)	4.5.33.4(N)	X				
13	CONDITIONING CHARGE	X	X	X	--	4.4.5(G)	X			X	X
14	CAPACITY DISCHARGE	X	X	X	Reqt (4)(N), 3.10.14(S)	4.5.34(N)	X			X	X
15	PROFILE DISCHARGE AT -18°C			X	3.10.32(N)	4.5.32(N)				X	
16	SHOCK (BASIC)	X			3.9(G), 3.10.16(G)	4.5.15(G)					
17	TEMPERATURE SHOCK		X		3.9(G), 3.10.17(G)	4.5.17(G)					
18	HUMIDITY		X		3.9(G), 3.10.15(S)	4.5.14(G)					
19	SALT FOG	X			3.9(G), 3.10.19(S)	4.5.19(G)					
20	TEMPERATURE RISE AND FLOAT		X		3.9(G), 3.10.9(S), 3.10.24(G)	4.5.24(G)					X

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TABLE I. Qualification and quality conformance inspection of PIN D8565/10-1. - continued

Test number	Examination and tests 6/	Qualification inspection					Conformance inspection				
		Sample number			Requirements paragraph 1/	Test paragraph 1/	Group A sample 100%	Group B sample		Group C sample	
		1	2	3				1	2	1	2
21	LIFE CYCLING	X		X	3.9(G), 3.10.23(G)	4.5.23(G)				X	
22	CONDITIONING CHARGE	X	X		---	4.4.5(G)			X	X	X
23	CONSTANT VOLTAGE DISCHARGE	X	X		3.9(G), 3.10.22(S)	4.5.22(G)			X4/	X4/	X4/
24	PHYSICAL INTEGRITY AT 85°C	X			3.9(G), 3.10.25(G)	4.5.25(S)					
25	DIELECTRIC STRENGTH AND INSULATION RESISTANCE	X	X	X	3.10.27(G)	4.5.27(G)				X	X
26	VIBRATION		X		3.9(G)	4.5.16(S)					
27	FINAL EXAMINATION	X	X	X	3.9(G), 3.10.27(G)	4.5.38(N)	X			X	X
28	INTERNAL PRESSURE		X		3.9(G), 3.10.26(G)	4.5.26(G)					
29	GAS BARRIER MATERIAL TEST			X	3.9(G), 3.10.9(S), 3.10.28(G)	4.5.28(G)					
30	SHELF LIFE	X	X		3.9(G), 3.10.30(G)	4.5.30(G)					
10	TEMPERATURE ACCURACY	X5/	X5/	X5/	3.10.33.2(N)	4.5.33.2(N)				X5/	X5/
11	SWITCH ACTIVATION	X5/	X5/	X5/	3.10.33.3(N)	4.5.33.3(N)				X5/	X5/
31	HARNESS ALKALI RESISTANCE			X5/	3.10.33.1(N)	4.5.33.1(N)				X5/	X5/
32	CORROSION RESISTANCE	3/	3/	3/	3.10.35(N)	4.5.35(N)					

1/ The notations in parentheses specify the source of the paragraph. "(G)" means the paragraph as shown in MIL-PRF-81757. "(S)" shows that the paragraph has been modified by this specification sheet. "(N)" indicates that a new paragraph or requirement has been added by MIL-PRF-81757/17.
2/ Conduct only the weight inspection.
3/ Metal parts that are representative of the metal parts contained in the three qualification samples shall be submitted.
4/ At -26 ± 2°C only.
5/ Remove the harness from the battery and then perform these tests in the sequence shown.
6/ Except for inspection numbers 2 through 11, inspections shall be conducted sequentially in the numerical order listed. The sequence of inspection numbers 2 through 11 may be interchanged.

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TABLE II. Qualification and quality conformance inspections PIN D8565/10-2.

Inspection number	Inspections[1/]	Qualification inspection Sample number 1 2 3	Refer to MIL-PRF-81757 and MIL-PRF-81757/17(AS)		Quality conformance inspection		
			Requirements paragraph 2/	Method of inspection paragraph 2/	Group A sample 100%		Group B sample 1 2
1	Visual and mechanical examination	X X X	Reqt (6) and (7)	4.5.33.6	X		X X
2	Dimensions and weight	X X	Reqt (7)	4.5.33.7	X		X X
3	Temperature accuracy	X X X	3.10.33.2	4.5.33.2	X		X X
4	Switch activation	X X X	3.10.33.3	4.5.33.3	X		X X
5	Switch contact resistance	X X X	3.10.33.4	4.5.33.4	X		X X
6	Harness alkali resistance	X	3.10.33.1	4.5.33.1			X
3	Temperature accuracy	X X X	3.10.33.2	4.5.33.2	X		X X
4	Switch activation	X X X	3.10.33.3	4.5.33.3	X		X X
5	Switch contact resistance	X X X	3.10.33.4	4.5.33.4	X		X X
7	Vibration	X	3.10.33.5	4.5.33.5			X

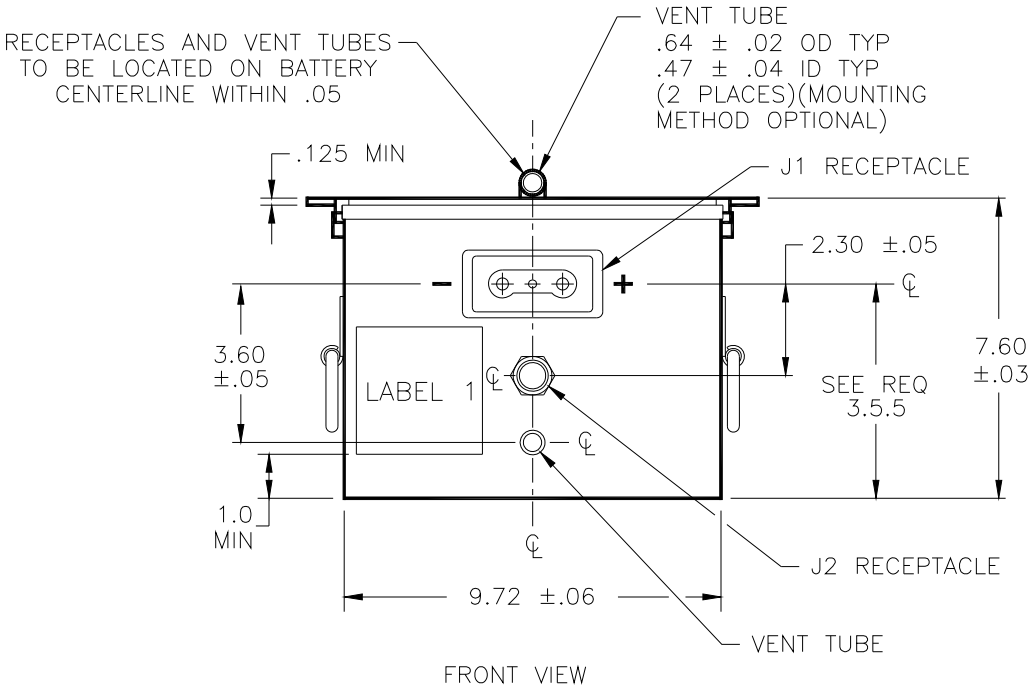
NOTES: 1/ Inspections shall be conducted sequentially in the numerical order listed.
2/ All paragraphs referenced are added by MIL-PRF-81757/17.

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TABLE III. Battery, cell, and component deleterious effects.

Affected part of the connector	Deleterious effects
External and internal parts	Cracking, pitting, chipping, scaling, blistering, corrosion (however, if a component [for example, a spring or receptacle pin] is made of 400-series stainless steel, then minor surface corrosion is permitted), or other deleterious effects
Electrical parts	Functional degradation
Plastic materials	Change more than 2.5 percent in weight or 2.0 percent in any major dimension (length, width, or height)
Identification markings	Smudging, smearing, chipping, crazing, or other deleterious effects

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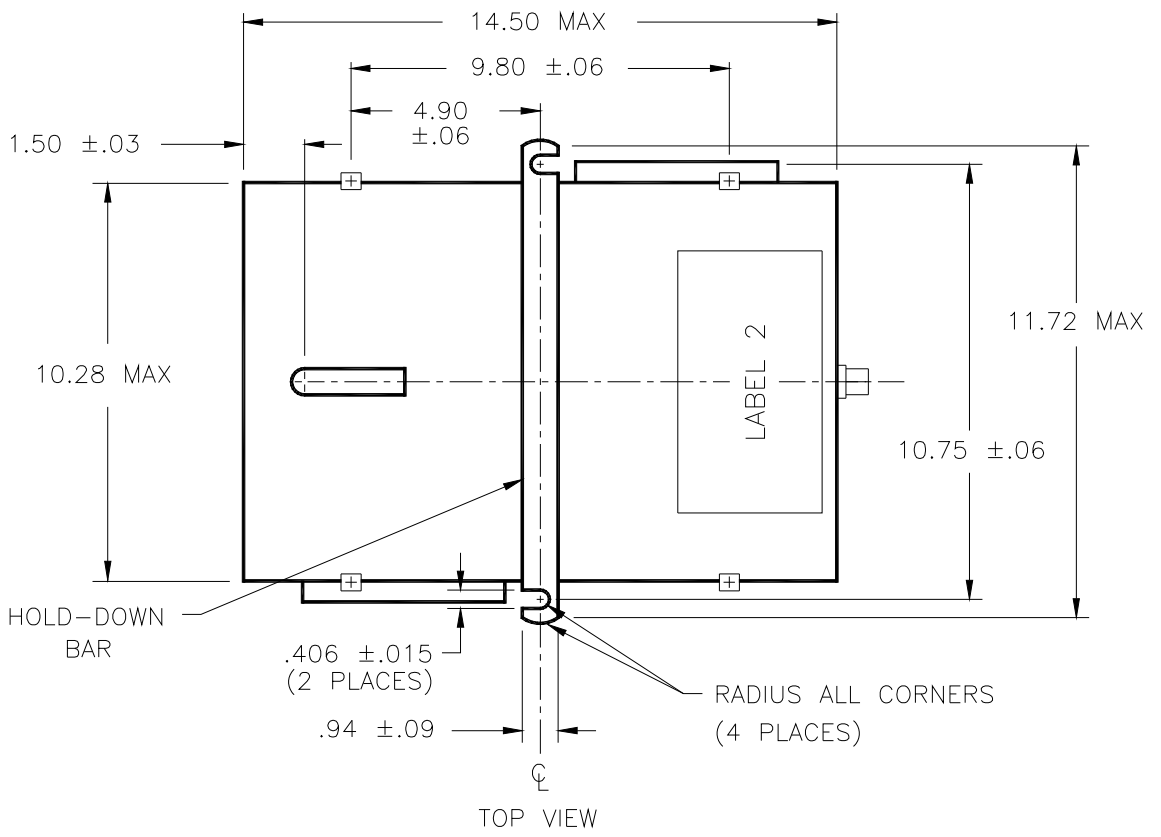


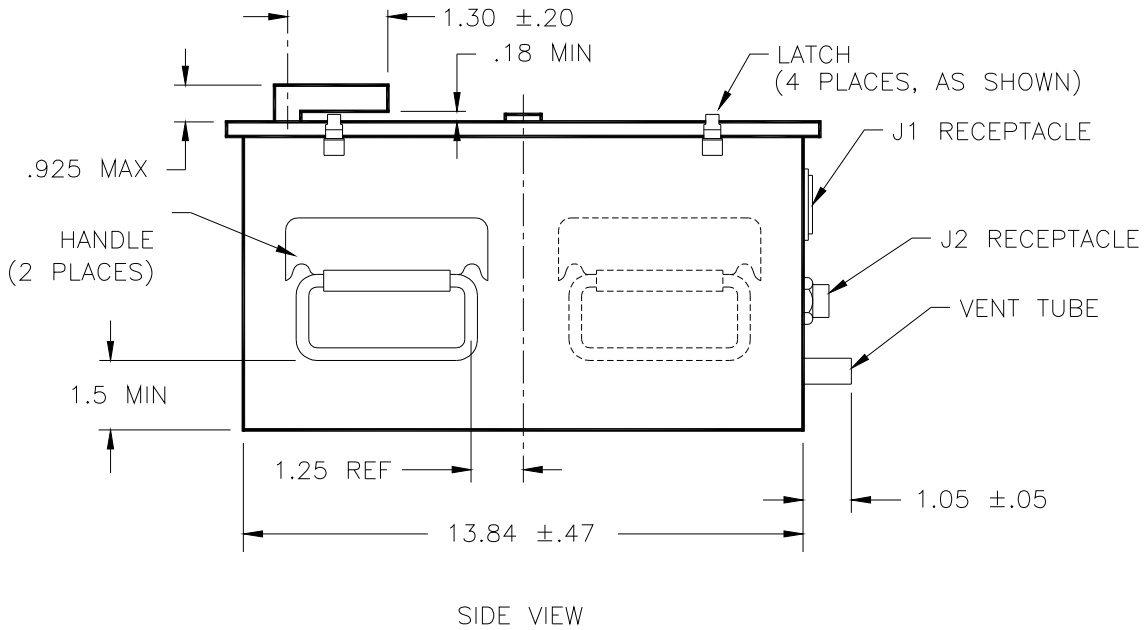
NOTES:

1. All dimensions are in inches.
2. See MIL-PRF-81757 requirement 3.5.5.

FIGURE 1. PIN D8565/10-1 dimensions.

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NOTES:

1. All dimensions are in inches.

FIGURE 1. PIN D8565/10-1 dimensions - Continued.

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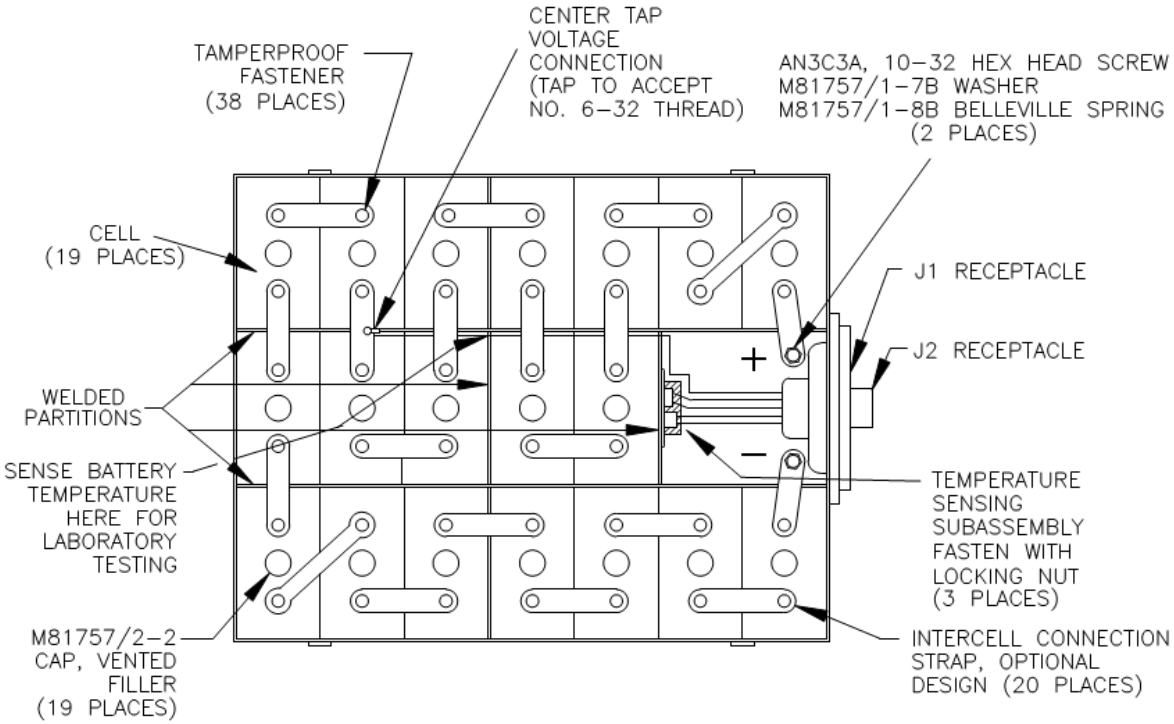


FIGURE 2. Layout of cells and sensors.

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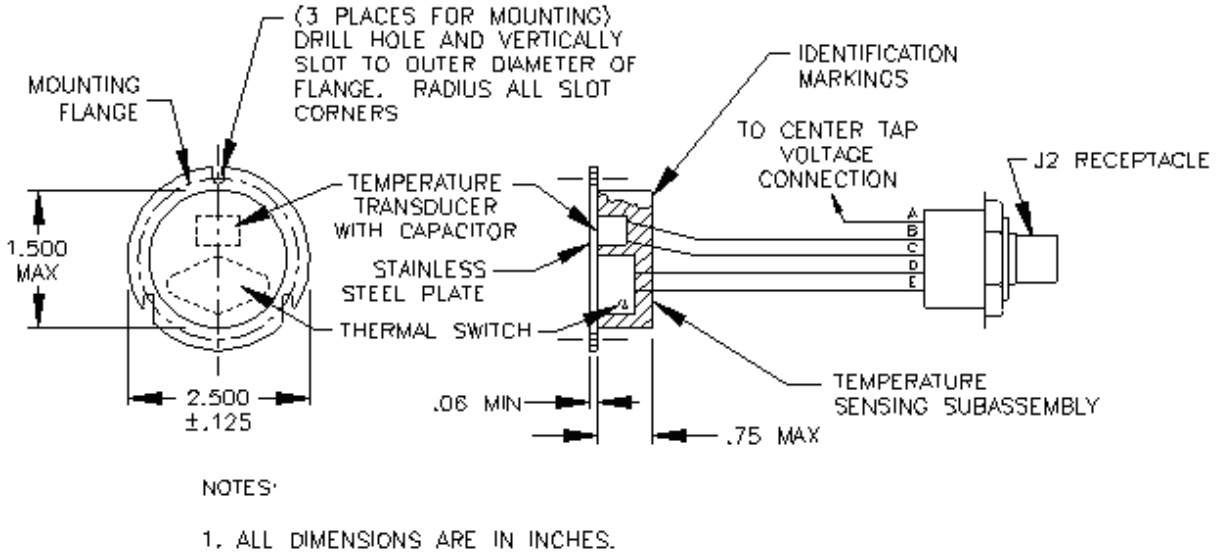


FIGURE 3. PIN D8565/10-2 sensor/connector assembly.

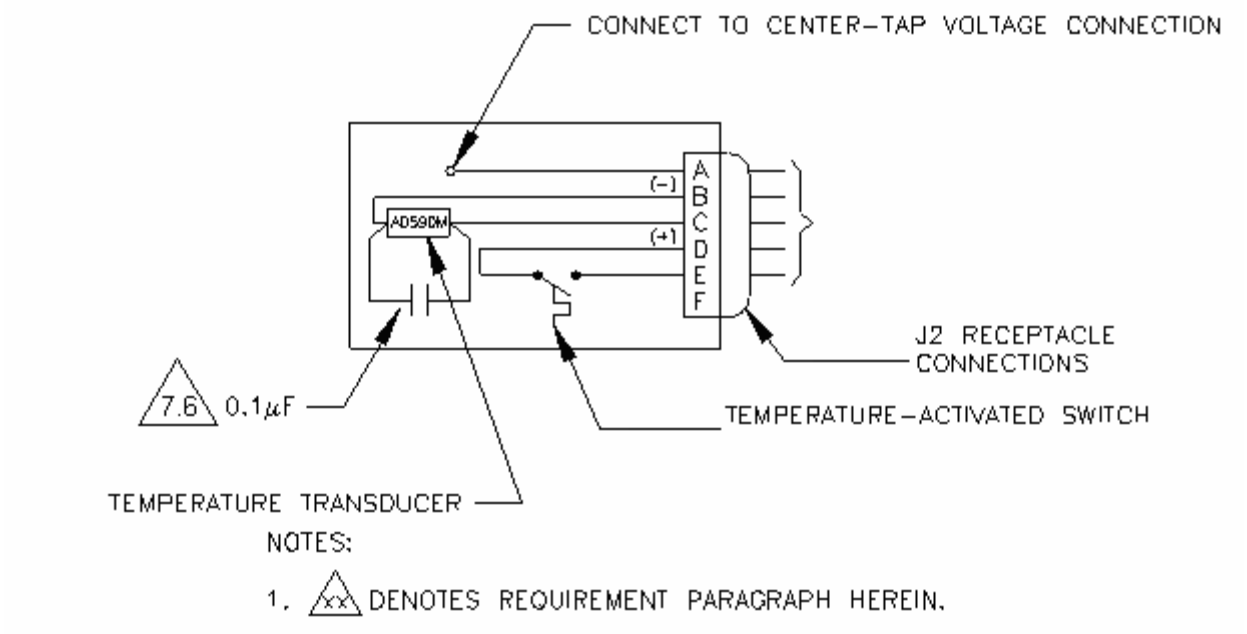


FIGURE 4. PIN D8565/10-2 electrical schematic.

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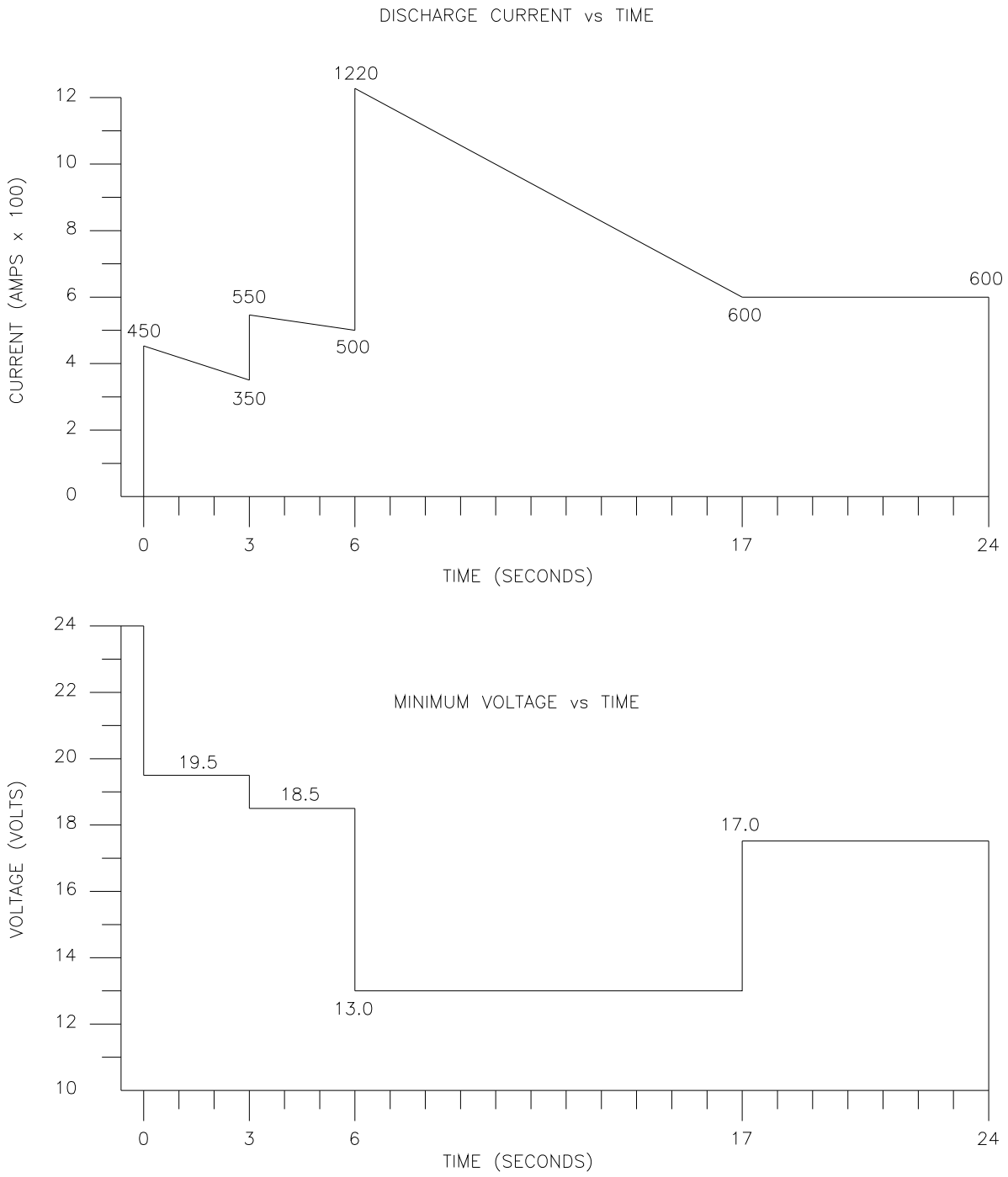
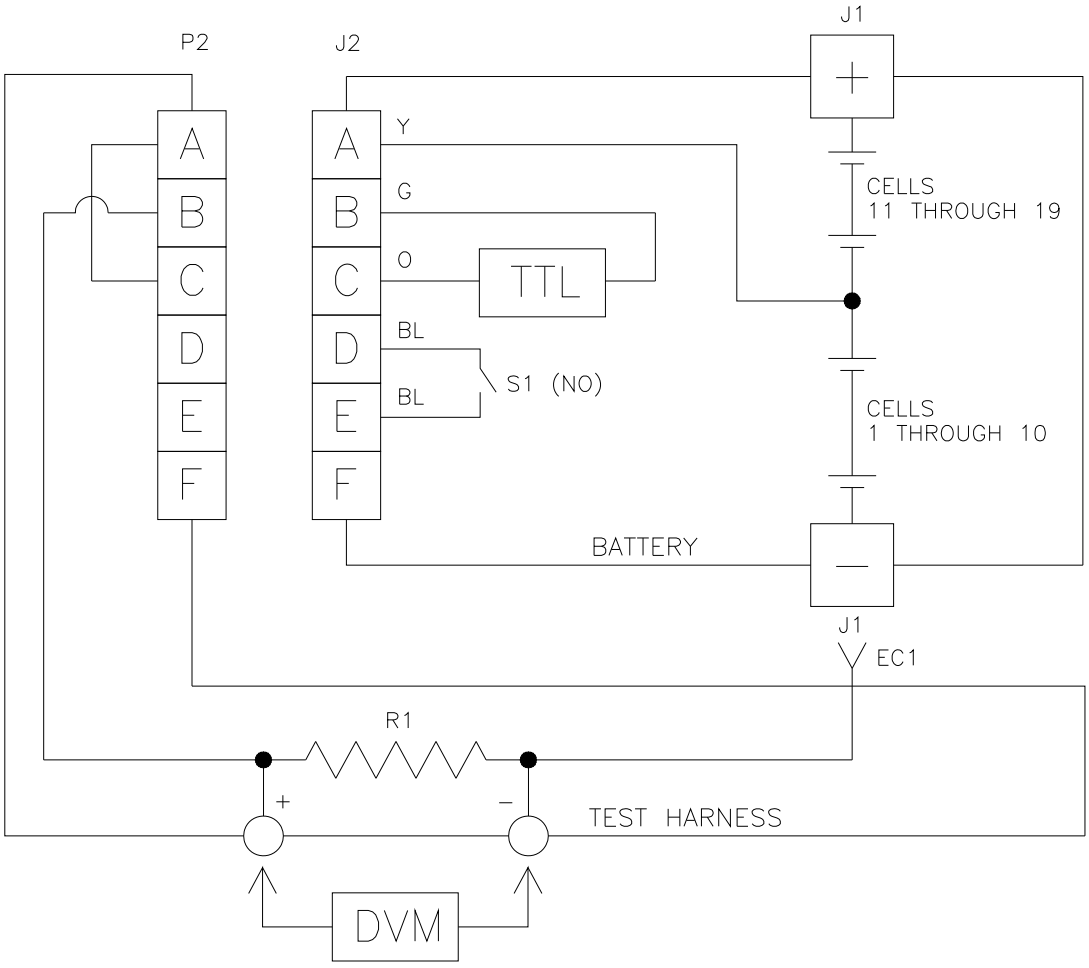


FIGURE 5. Profile discharge at -18ºC.

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BATTERY

J1 – RECEPTACLE (SPECIFIED HEREIN)
J2 – RECEPTACLE (SPECIFIED HEREIN)
S1 – THERMOSWITCH:
CLOSES ON A TEMPERATURE RISE AT 74 ± 3°C
REOPENS ON A TEMPERATURE FALL DIFFERENTIAL OF 6°C MAXIMUM
TTL – TRANSISTOR–TRANSISTOR LOGIC DEVICE AD590M(4.0 TO 30.0 VOLTS DC REQUIRED)

TEST HARNESS

DVM – DIGITAL VOLTMETER
EC1 – ELECTRICAL CLIP W/INSULATOR
P2 – (MS3476L10–6S) MATES W/RECEPTACLE J2
R1 – 10,000 OHM, 0.1 PERCENT, 1/10 WATT PRECISION RESISTOR

NOTES: 1. CELLS ARE NUMBERED NEGATIVE TO POSITIVE AS SCANNED BY THE NBC–1A CHARGER/ANALYZER (REFERENCE TECHNICAL MANUAL NAVAIR17–15BAD–1)

2. ABBREVIATIONS USED IN THIS FIGURE ARE NORMALLY OPEN (NO), AND PART OR IDENTIFYING NUMBER (PIN).

FIGURE 6. Circuit for conducting PIN function tests.

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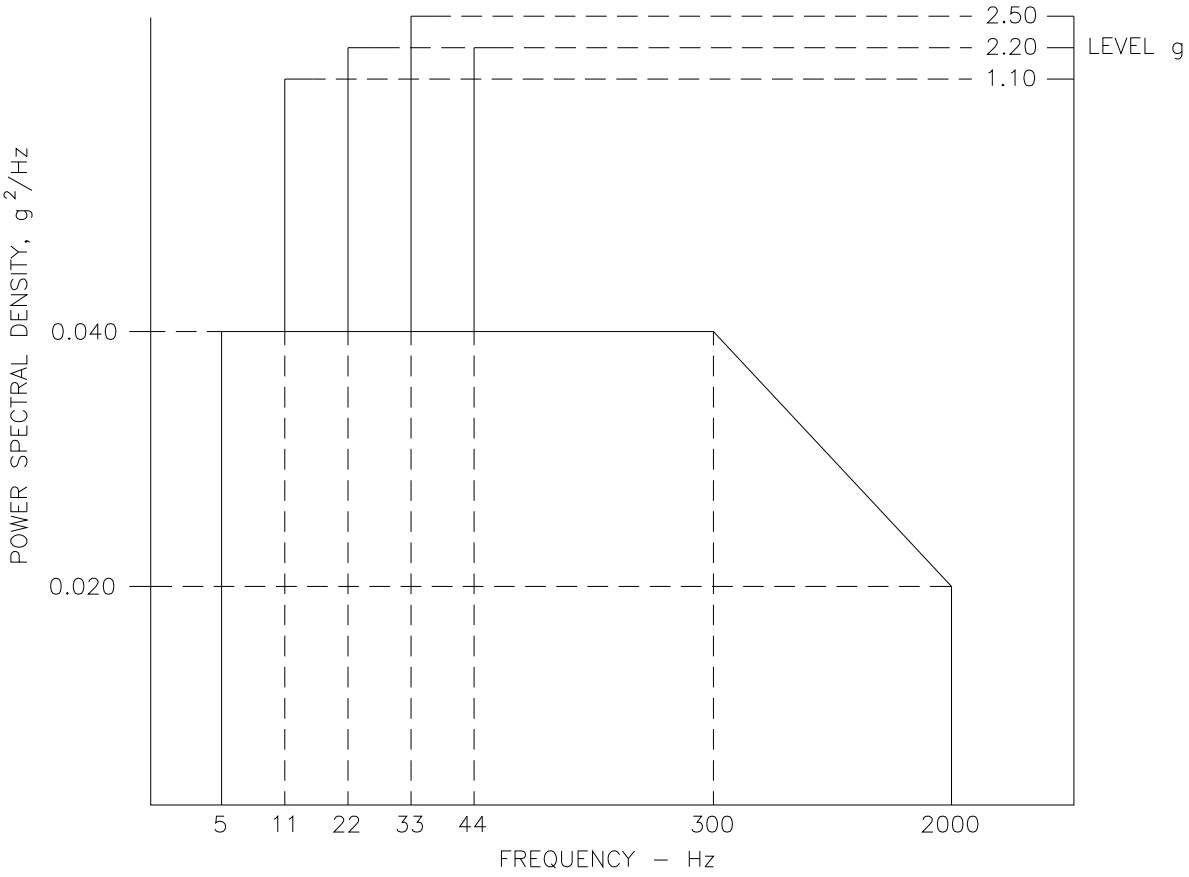


FIGURE 7. Vibration profile.

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CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing Activity:
Navy - AS

Review Activity:
Navy - SH

Agent:
Navy - SH

(Project 6140-2008-007)

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